FMGT 3510 – Finance 1
Fall 2026

Chapter 3

Financial Statement Analysis (FSA)

Overview of Financial Statements Analysis (1 of 2)

Financial Statements Analysis (FSA) allows us to **evaluate and compare** the financial position and financial performance of a firm. It can also help us **forecast** future earnings, dividends and free cash flow.

Many online sources provide FSA of public companies such as **S&P Capital IQ**
S&P Capital iQ if one of the most used professional database. It is available to BCIT students through the Library's website.

The starting point of any financial statements analysis is to examine **the statement of cash flows.**

- Are **operations providing or using** cash flow? What is the **trend** in cash from operations?
- Are there large amounts of cash used in **investing** activities?
- Does cash from **financing** activities show cash coming in to the firm or being paid out to investors? Which type of investors (debt or equity) are receiving or paying cash?

The **next step** would be to calculate **relevant financial ratios**, and the **Free Cash Flow,** and **analyze** their **trends.**

Overview of Financial Statements Analysis (2 of 2)

In this chapter we will study the **main tools used by financial analysts**:

- **Comparative financial statements**: the current financial position and the operating results for the current year are compared to previous years.
- **Common size financial statements**: items on the balance sheet and income statement are expressed as a percentage of a total (total assets for the balance sheet and total revenue for the income statement).
- **Financial ratios**: the examination of **relationships between certain key items** on the financial statements.

Financial analysts normally use two approaches:

Inward approach: examines firm's performance overtime (trend analysis) against past record.

Outward approach: compares firm's performance against other companies in the same industry.

Outward approach must be used **cautiously** as regional differences, size differences, differences in the application of accounting policies and poor category differentiation often make it **difficult to find truly comparable companies** within an industry.

How can we assess the performance of a firm? (1 of 2)

The performance of a firm can be evaluated from different perspectives.

Liquidity	The firm's ability to meet sudden cash requirements and to **meet its short-term financial obligations** as they come due.
Productivity *asset management*	The firm's **ability to put its assets to good use**. Is the firm generating sufficient sales from its various investments?
Profitability	The firm's ability to generate **sufficient profits** in **comparison** to its level of **sales** and **invested capital**.
Leverage	The degree to which the firm is financed with borrowed money (debt). How much **financial risk** is the firm exposed to considering its **capital structure** (mix of interest paying debt and equity financing used) ?
Valuation	The value that investors assign to the firm. This can be easily measured for public companies. It is **more difficult to establish the value of private companies** as their shares are not traded in the capital markets.

How can we assess the performance of a firm? (2 of 2)

Which perspective is the most important? It depends on who you are.

Suppliers and banks (short-term lenders):

Liquidity, Leverage

Shareholders:

Profitability, valuation, leverage

Long-term creditors:

Leverage, liquidity

Management:

Valuation, productivity, profitability, leverage, etc.

We will calculate Ellington Ltd.'s financial ratios for 2025 using the following financial statements (1 of 2)

Ellington Ltd. Balance Sheet as at December 31st

in thousands of $	2025	2024		2025	2024
Assets			**Liabilities and Equity**		
Cash	98	84	Accounts payable	344	312
Accounts receivable	188	165	Short term bank loans	196	231.
Inventories	422	393	Total current liabilities	540	543
Total current assets	708	642			
			Long-term debt	457	531
Net Fixed assets	2,880	2,731	Shareholders' equity		
			Common stock	550	500
			Retained earnings	2,041	1,799
			Total equity	2,591	2,299
Total assets	3,588	3,373	Total liabilities & equity	3,588	3,373
			Common shares oustanding	52,500	50,000
			Common share price today	$51.00	

We will calculate Ellington Ltd.'s financial ratios for 2025 using the following financial statements (2 of 2)

Ellington ltd. Income statement for the years		
in thousands of $	**2025**	**2024**
Sales	2,311	2,075
COGS	1,344	1,295
Depreciation	276	249
EBIT	691	531
Interest expense	141	129
EBT	550	402
Taxes	187	162
Net Income	363	240
Dividends	*121*	*120*
Addition to Retained Earnings	*242*	*120*
Earnings per Share (EPS)	$6.91	$4.80
Dividend per share (DPS)	$2.30	$2.40
DPS = total dividends/number of c/shares		
COGS includes $25,000 in lease payments		

Liquidity ratio

Ratio	What it measures	Formula	Example
Current ratio	$'s of current assets per $ of current liabilities	$\frac{Current\ Assets}{Current\ Liabilites}$	**708 / 540 = 1.31 x**
Quick ratio	$'s of liquid current assets per $ of current liabilities	$\frac{Current\ Assets - Inventories}{Current\ Liabilites}$	**(708 – 422) / 540 = 0.53 x**

In comparison to industry averages, a **low current ratio** may indicate a **liquidity shortage**, which could lead to future cash flow problems.

An **excessively high current ratio** may indicate **over investment in short-term assets** and/or **underutilization of trade credit or short-term borrowing facilities**.

A **rising trend in the current ratio** and a **falling trend in the quick ratio** may indicate **rising inventory levels** and/or **slower inventory turnover**.

Productivity (Asset Management) analysis (1 of 2)

Ratio[2]	What it measures	Formula	Example
Inventory turnover ratio	Number of times inventory must be replaced during the year	$\frac{COGS}{Inventories}$	**1,344 / 422 = 3.18 x**
Days sales outstanding (DSO)	Average length of time between sale and receiving cash	$\frac{Receivables}{\left(\frac{Annual\ Sales}{365}\right)}$	**188 / (2,311 / 365) = 29.7 days**
Average payables period (APP)	Average number of days of COGS not yet paid	$\frac{Payables}{\left(\frac{Annual\ COGS}{365}\right)}$	**344 / (1,344/365) = 93.4 days**
Fixed assets turnover ratio	$ of sales per $ of fixed assets	$\frac{Sales}{Net\ Fixed\ Assets}$	**2,311 / 2,880 = 0.80 x**
Total assets turnover ratio	$ of sales per $ of total assets	$\frac{Sales}{Total\ Assets}$	**2,311 / 3,588 = 0.644 x**

[2] Year end balances used in calculations; average balances, for balance sheet items only, are also sometimes used.

Productivity (Asset Management) analysis (2 of 2)

Normally, a firm is looking for **high asset turnovers** and a **low payables turnover**.

However, **excessively high asset turnovers** can indicate an **unsustainable level of operations or an insufficient investment** in certain assets.

Extremely **low asset turnovers may** indicate that the firm's **investment in certain assets is too high or** that **management is doing a poor job**.

Profitability analysis (1 of 2)

Ratio	What it measures	Formula	Example
Net profit margin	Net income relative to sales	$\frac{Net\ income\ available\ to\ common\ shareholders}{Sales}$	363 / 2,311 **= 15.71%**
Gross profit margin	Gross profit per $ of sales before other expenses	$\frac{Sales - Cost\ of\ Goods\ Sold}{Sales}$	(2,311 – 1,344) / 2,311 **= 41.84%**
Basic earnings power	Earning power of assets before taxes and impact of leverage	$\frac{EBIT}{Total\ Assets}$	691 / 3,588 **= 19.26%**
Operating profit margin	Profit before impact of interest and taxes	$\frac{EBIT}{Sales}$	691 / 2,311 **= 29.90%**

Note: Net income available to common shareholders ("NIATCS") is net income **after deducting any preferred share dividends**.

Profitability analysis (2 of 2)

Ratio	What it measures	Formula	Example
Return on assets (ROA)	Net income available to common relative to total assets	$\frac{Net\ income\ available\ to\ common\ shareholders}{Total\ Assets}$	**363 / 3,588 = 10.12%**
Return on common equity (ROE)	Net income available to common relative to common equity	$\frac{Net\ income\ available\ to\ common\ shareholders}{Common\ Equity}$	**363 / 2,591 = 14.01%**
Return on invested capital (ROIC)	Ratio of NOPAT to Operating Capital (see Ch 2)[1]	$\frac{NOPAT}{Operating\ Capital}$	**691 (1 - 0.34) / 3,244 = 14.06%** ***550(T) = 187*** ***Tax rate = T = 187/550 = 0.34***

Generally, the **higher the profitability ratios**, the **better**.

Low profitability may indicate **poor cost control** and/or a **sales mix** that is skewed toward **lower margin** products.

Large variations in the **gross profit margin** from year-to-year may indicate **inventory valuation problems.**

[1] Operating Capital = NOWC + Operating Long-term Assets = (708 – 344) + 2,880 = 3,244
NOPAT = EBIT(1 – tax rate) = 691(1 – 0.34) = 456

Leverage (debt management) (1 of 2)

Ratio	What it measures	Formula	Example
Debt ratio	% of capital provided by short term and long-term debt	$\frac{Total\ Debt}{Total\ Assets}$	**(196 + 457)/ 3,588 = 18.2%**
Times interest earned ratio (TIE)	$ of operating earnings available to pay each $ of interest expense	$\frac{EBIT}{Interest\ Expense}$	**691 / 141 = 4.9 x**
EBITDA coverage ratio	$ of operating cash flow available to pay each $ of fixed charges	$\frac{EBITDA + Lease\ Payments}{Interest + Principal\ Payments + Lease\ payments}$	**(691 +276 +25) / 141 + 25 = 6 x**
Debt to equity	$ of debt for each $ of equity	$\frac{Total\ Debt}{Shareholders'\ Equity}$	**(196 + 457) / 2,591 = 0.25**
Assets-to-equity (Equity multiplier)	$ of assets for each $ of equity	$\frac{Total\ Assets}{Shareholders'\ Equity}$	**3,588 / 2,591 = 1.385**

Note: principal payments correspond to sinking funds payments. None here.

Leverage (debt management) (2 of 2)

The leverage or debt management ratios help us asses the firm's **degree of financial leverage and financial risk.**

A firm with excessively high levels of debt may be considered too risky and be penalized with **poor credit ratings** and **high required rates of return on its debt and equity**.

A firm with too low levels of debt may be considered **under levered** and its value may not be maximized.

TECK Resources: Debt Ratio = 25.5%

CN Rail: Debt Ratio = 51%

(Source: S&P Capital iQ, June 2026)

Valuation ratios (1 of 3)

Book Value per Share (BVPS)

Book Value per Share, is also called Book Value of Common Equity.

Book value of common equity, from the balance sheet, in $2,591,000. We then divide this amount by the number of common shares outstanding:

$2,591,000 / 52,500 = $49.35 = Book Value per Share ("BVPS") at the end of 2025.

Cash flow per Share (CFPS)

CFPS is defined as **Net Cash Flow (Net CF)** divided by the number of common shares outstanding.

NET CF is equal to **net income plus depreciation and amortization.** Not to be confused with the Operating CF which is equal to NOPAT + depreciation and amortization.

NET CF = 363,000 + 276,000 = $639,000

CFPS = $639,000 / 52,500 = $12.17

Valuation ratios (2 of 3)

Ratio	What it measures	Formula	Example
Price / earnings ratio (P/E)	Amount investors pay per $ of EPS	$\frac{Common\ Share\ Price}{EPS}$	***EPS = 363,000 / 52,500 = 6.91$*** **51 / 6.91 = 7.38 x**
Price / cash flow ratio	Amount investors pay per $ of cash flow per share	$\frac{Common\ Share\ Price}{CFPS}$	**51 / 12.17 = 4.2 x**
Market / book ratio	How much investors pay per $ of book value per share	$\frac{Common\ Share\ Price}{BVPS}$	**51 / 49.35 = 1.033 x**
Dividend yield	Return provided by dividends per share	$\frac{Dividend\ per\ Common\ Share}{Common\ Share\ Price}$	***121,000 / 52,500 = 2.3$*** **2.3 / 51 = 4.51%**

Valuation ratios (3 of 3)

A firm may have a **high P/E ratio** due to **excessively low earnings** and/or **high expected growth**. Investors may be very optimistic about the firm's prospects.

Dividend yield is usually tied to rates of growth (see examples below).

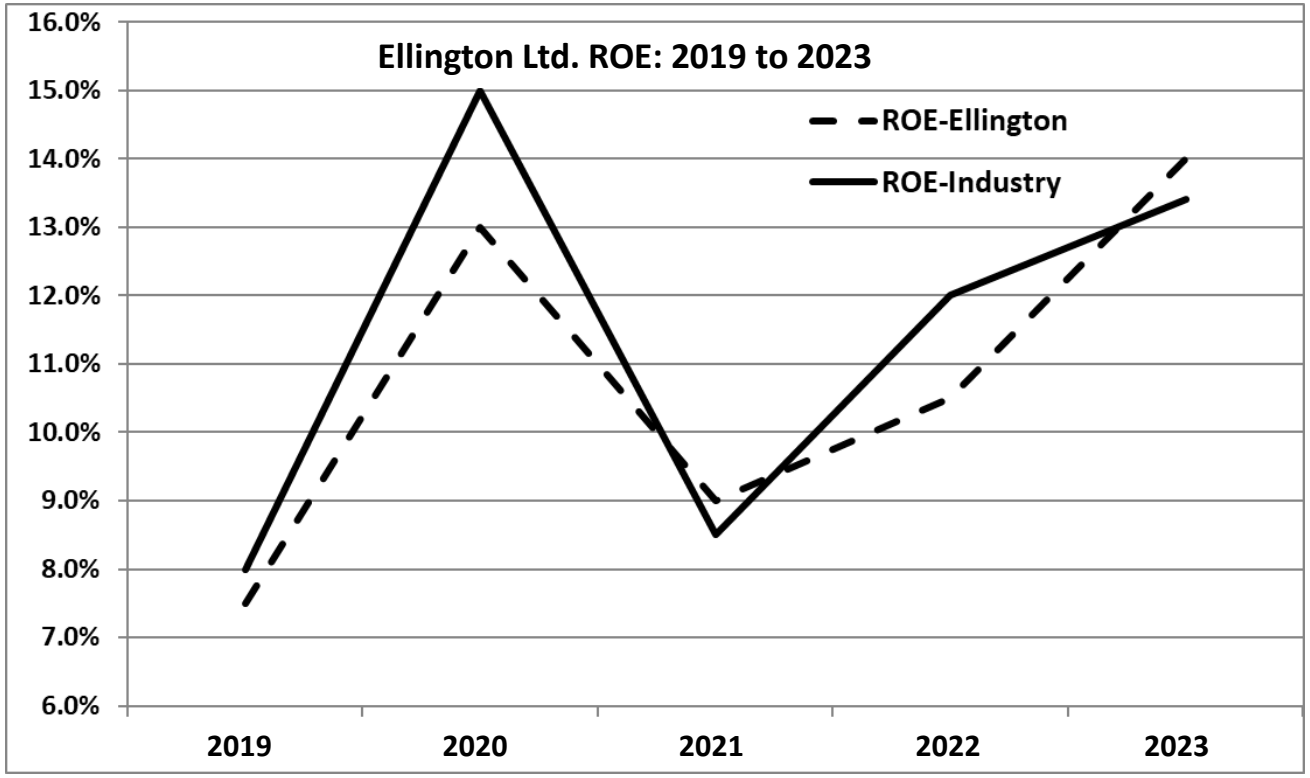
High-growth firms typically reinvest most of their earnings instead of paying them out as dividends, resulting in low yields.

Slow-growth firms in stable industries typically pay out a higher percentage of their earnings and have a higher yield.

examples	Palantir	SpaceX	Apple	RBC
DPS	USD 0	USD 0	USD 1	CAD 6.25
Dividend Payout ratio	0%	0%	13.8%	43.2%
Dividend yield	0%	0%	0.34%	2.28%
Stock price	USD117.5	USD 139.63	USD 313.45	CAD 288.11
EPS	USD 1.2	- USD 1.28	USD 8.73	CAD 15.64
PE ratio	150.42x	PE n.a.	35.9x	18.72x

Trend analysis, common size analysis and percent change analysis (1 of 3)

Trend analysis involves plotting a ratio over time. This figure shows ROE over a five-year period



Trend analysis, common size analysis and percent change analysis (2 of 3)

Common size statements (also known as **vertical analysis**) make it easier to compare changes in income statements and balance sheets over time.

- **Balance sheet items are divided by total assets**
- Income statement items are divided by sales
- Percent change analysis calculates the % growth in balance sheet and income statement items over time.

Ellington Ltd. Balance Sheet as at December 31st

in thousands of $	2025	2024		2025	2024
Assets			**Liabilities and Equity**		
Cash	2.7%	2.5%	Accounts payable	9.6%	9.2%
Accounts receivable	5.2%	4.9%	Short term bank loans	5.5%	6.8%
Inventories	11.8%	11.7%	Total current liabilities	15.1%	16.1%
Total current assets	19.7%	19.0%			
			Long-term debt	12.7%	15.7%
Net Fixed assets	80.3%	81.0%	Shareholders' equity		
			Common stock	15.3%	14.8%
			Retained earnings	56.9%	53.3%
			Total equity	72.2%	68.2%
Total assets	100.0%	100.0%	Total liabilities & equity	100.0%	100.0%

Accounts receivable have increased as a % of total assets.

Trend analysis, common size analysis and percent change analysis (3 of 3)

Common size statements (also known as **vertical analysis**) make it easier to compare changes in income statements and balance sheets over time.

- Balance sheet items are divided by total assets
- **Income statement items are divided by sales**
- Percent change analysis calculates the % growth in balance sheet and income statement items over time.

Ellington ltd. Income statement for the years		
in thousands of $	**2025**	**2024**
Sales	100.0%	100.0%
COGS	58.2%	62.4%
Depreciation	11.9%	12.0%
EBIT	29.9%	25.6%
Interest expense	6.1%	6.2%
EBT	23.8%	19.4%
Taxes	8.1%	7.8%
Net Income	15.7%	11.6%
Dividends	*5.2%*	*5.8%*
Addition to Retained Earnings	*10.5%*	*5.8%*

COGS has dropped as a % of sales

DuPont analysis (1 of 3)

The **DuPont system** integrates three performance perspectives, ***profitability, productivity and leverage,*** to help us understand the firm's **financial strategy.**

DuPont analysis focuses on **return on equity (ROE)** which is calculated by combining the following three ratios:

Net profit margin = NI/Sales
How many cents of after-tax profit does each dollar of sales generate ?

Total Asset Turnover = Sales / Total Assets
How many dollars of sales does each dollar of assets generate ?

Equity multiplier = Total Assets / Equity
How many dollars of assets does the firm have for every dollar equity capital?

DuPont analysis (2 of 3)

$$ROE = Profitability \times Productivity \times Leverage$$

$$ROE = \frac{Net\ Income}{Shareholders'\ Equity}$$

$$ROE = \frac{Net\ Income}{Sales} \times \frac{Sales}{Total\ Assets} \times \frac{Total\ Assets}{Shareholders'\ Equity}$$

Net profit margin

Total Asset Turnover

Equity multiplier

ALSO,

$$ROA = \frac{Net\ Income}{Total\ Assets} = \frac{Net\ income}{Sales} \times \frac{Sales}{Total\ Assets}$$

$$ROE = ROA \times \frac{Total\ Assets}{Shareholders'\ Equity}$$

DuPont analysis – Ellington example (3 of 3)

$$ROE = \frac{Net\ Income}{Sales} \times \frac{Sales}{Total\ Assets} \times \frac{Total\ Assets}{Shareholders'\ Equity}$$

			Net profit margin		Total Assets Turnover		Equity multiplier
ROE(2025)	14.01%	=	15.71%	x	0.64	x	1.385
ROE(2024)	10.44%	=	11.57%	x	0.62	x	1.467

Limitations of financial statement analysis

Amounts on the balance sheet are generally based on **historic cost** and not market value.

- Ratios that involve various balance sheet accounts may not be very meaningful if there is a big difference between market value and book value.
- They may have very little direct relationship to the firm's ability to generate cash flow.

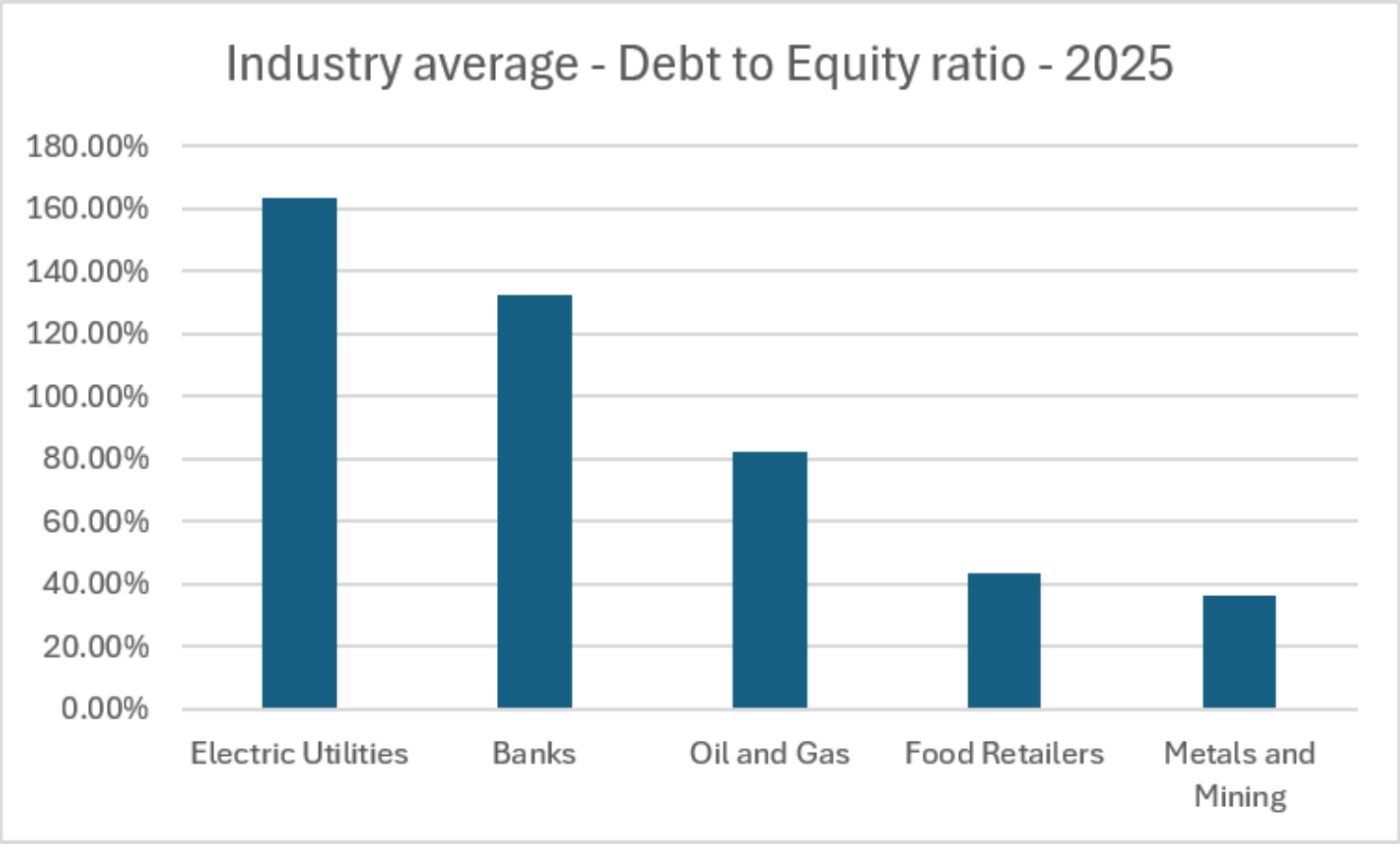
The use of **different accounting policies**. This may make comparisons between different companies difficult.

Firms may use **"window dressing"** to make their financial statements look better.

Many firms have **highly summarized** financial statements with few details. This makes it difficult to perform FSA.

High levels of **inflation or deflation** can cause distortions in the firm's financial results. It is **difficult to make industry comparisons** when the firm operates businesses that cross many different industries **(Conglomerate).**

Example of industry specificities



Source: S&P Capital iQ